

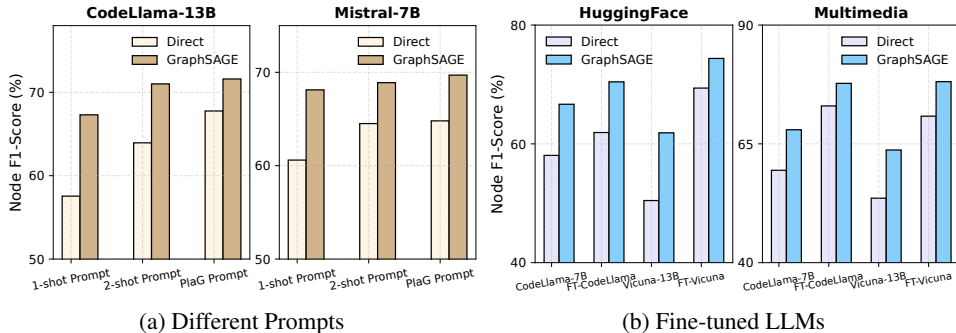


Figure 3: **Orthogonal Effectiveness to both Improved Prompts and Fine-tuned LLMs**

the results indicate that (1) LLMs’ performance is vulnerable to the scale of task graphs; (2) The performance gain of the proposed method increases with a larger task graph.

5 Improved Prompts and Fine-tuned LLMs

In this subsection, we show that the proposed method is orthogonal to two dominant methods, i.e., prompt engineering and fine-tuning.

Orthogonal to Improved Prompts: We investigate GNN’s effectiveness when applied to improved prompt templates, i.e., strategically designed prompts that enhance the task planning abilities of LLMs. Specifically, we consider two types of prompts: **(1) In-context Learning with Increased**